The KCP Limited

Thermal Stress

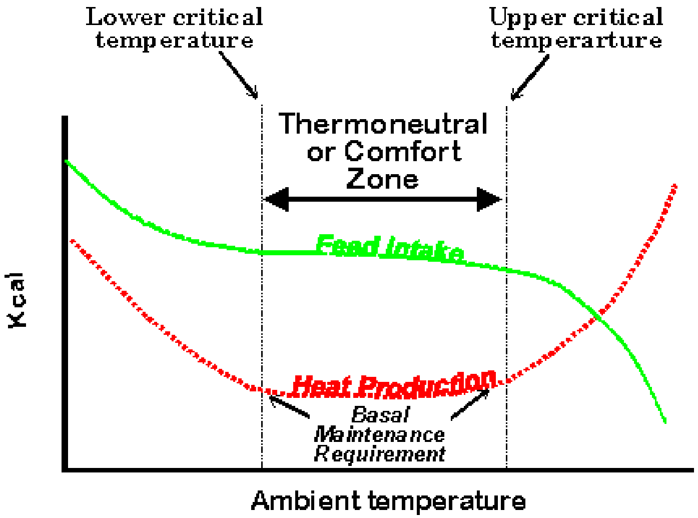
Introduction

- Thermal stress is a significant physical agent in many workplace environments.
  - Just considering routine out-of-doors work, air temperatures ranging from -20°F to 110°F can reasonably be expected in many places throughout the United States.

Degrees of Thermal Stress

Conceptually, work can occur in one of five zones along the continuum of thermal stress:

- **heat health risk zone**
- **heat discomfort zone**
  - physiological adaptions great
  - behavioral adaptations modest-to-great
- **comfort zone**
  - physiological adaptions modest
  - behavioral adaptations minor
- **cold discomfort zone**
  - physiological demands great
  - behavioral adaptations modest-to-great
- **cold health risk zone**



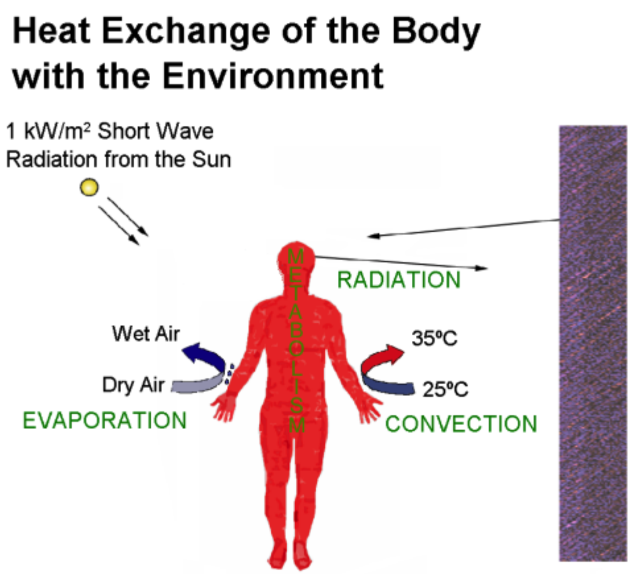
Model of Thermal Balance

- Three factors influence the degree of thermal stress:
  - climatic factors
    - air temperature, relative humidity, wind chill
  - work demands
    - substantial contributor to heat gain
  - clothing
    - insulation (resistance to heat flow)
    - permeability (water vapor movement)
    - ventilation (air movement)
      - chimney effect + bellows effect

Model of Thermal Balance (simplified)

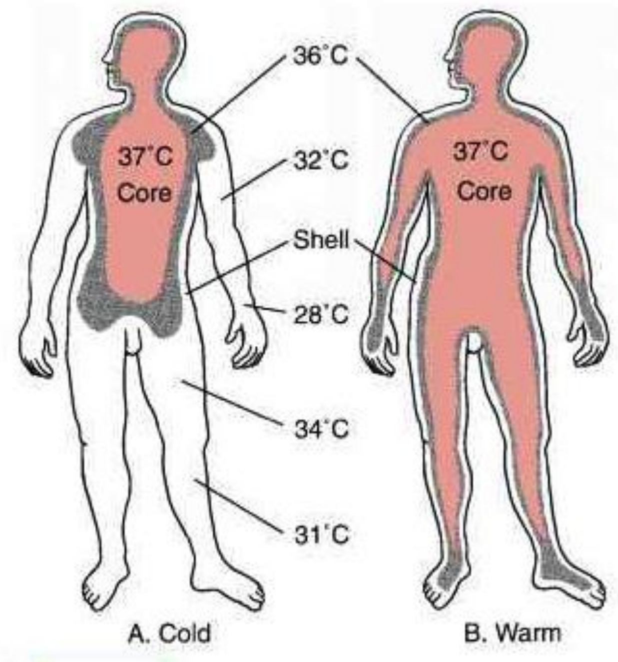
- S = M + R + C – E

  where: S = heat storage rate (gain [+] or loss [-])
  M = metabolic rate (internal heat generation [+])
  R = radiant heat exchange rate (gain [+] or loss [-])
  C = convective heat exchange rate (gain [+] or loss [-])
  E = evaporative heat loss (always loss [-])



Heat Stress

- A combination of factors tends to increase body temperature, heart rate, and sweating.
  - The physiological adaptations to heat stress are known as **heat strain**.
  - core body temperature
  - heart rate
  - sweat rate
    - acclimation
    - adaptive working (pacing)



Recognition of Heat Stress

- Heat stress in the workplace can be recognized in terms of:
  - workplace risk factors
    - hot or humid environments
    - high work demands
    - protective clothing requirements
  - effects on workers
    - low-end: worker behaviors (shedding clothes, work pacing)
    - middle: redness of face/skin, profuse sweating, rapid breathing
    - high-end: heat-related disorders (nausea, cramps, heat exhaustion)

Heat-Related Disorders

- Heat-related disorders are the manifestations of over-exposure to heat stress.
  - see Table 12–A in text for information on:
    - signs a trained observer may see
    - symptoms a worker may report
    - likely causes of the disorder
    - first aid
    - steps for prevention
  - physiological markers
    - measuring temperature (100.4°F)
    - measuring heart rate (< 110 *bpm* average or < 160 *bpm*)
    - measuring water loss (> 1.5% loss of body weight)

Worker Behaviors

- Heat stress not only induces physiological changes, but also affects behavior:
  - actions that reduce exposures
    - adjusting clothing to increase evaporative losses
    - slowing work rate
    - taking small breaks
    - taking shortcuts in work methods
  - changes in attitudes
    - irritability
    - low morale
    - absenteeism
  - unsafe acts
    - increased number of errors
    - increased number of machine breakdowns
    - taking short cuts in work methods

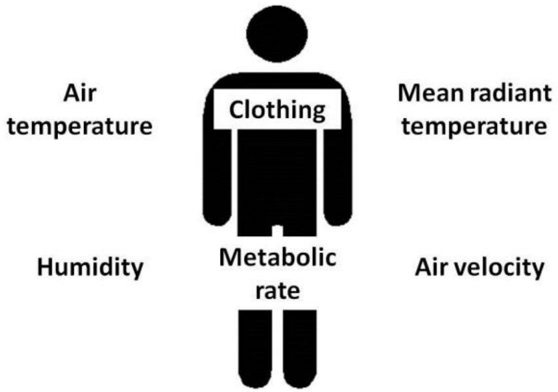
Evaluation of Heat Stress

- The World Health Organization set the tone for worker protection in 1969.
  - forms the basis for NIOSH and ACGIH heat stress guidelines
    - long-duration exposures
      - core temperatures should not exceed 100.4°F
    - short-duration exposures (with recovery periods)
      - core temperatures should not exceed 102.2°F
    - average heart rate
      - not over 110 bpm

ACGIH® TLV® for Heat Stress

- Recognizes the difference between eight-hour and short-duration exposures.
  - assessment of metabolic rate
    - basal metabolism
    - posture
    - activity (based on body movement)
    - horizontal travel
    - vertical travel
  - assessment of environmental conditions
    - air temperature
    - humidity
    - air speed
    - average temperature
  - clothing adjustment factor (CAF)

	Average (watts)	Range (watts)
Basal metabolism (*B*)	70	
Posture Metabolism (*P*)		
sitting	20	
standing	40	
walking	170	140-210
Activity Metabolism (*A*)		
hand (light)	30	15-85
hand (heavy)	65	
one-arm (light)	70	50-175
one-arm (heavy)	120	
both-arms (light)	105	50-175
both-arms (heavy)	175	
whole-body (light)	245	175-1050
whole-body (mod)	350	
whole-body (heavy)	490	
whole-body (very heavy)	630	



Control of Heat Stress

- The control of heat stress centers around the causes and their resulting physiological strain.
  - Takes the form of:
    - general controls
      - applicable to all heat-related jobs
    - specific controls
      - evaluated and selected based on constraints of the working conditions

Control of Heat Stress (cont.)

- General Controls
  - training
    - pre-placement training
    - periodic training
  - heat stress hygiene practices
    - fluid replacement
    - self-determination
    - diet
    - lifestyle
    - health status
    - acclimation
  - medical surveillance
  - evaluation of risk
  - response to heat-related disorders
  - emergency plan

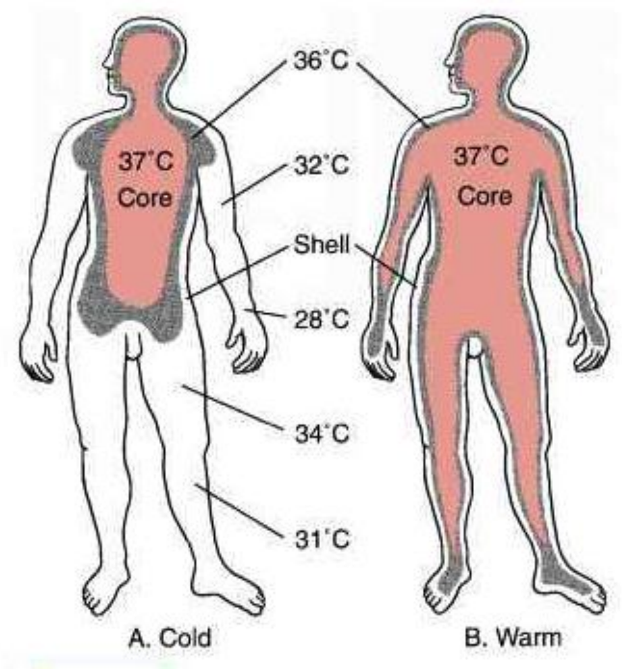
Control of Heat Stress (cont.)

- Specific Controls
  - engineering controls
    - reducing physical work demands
    - adjusting clothing requirements
    - reducing external heat gain
    - enhancing external heat loss

  - administrative controls
    - acclimation
    - work-pacing
    - work-sharing
    - work scheduling
    - pre-determined work times
    - self-determination
    - personal monitoring
  - personal protection
    - circulating air systems
    - circulating water systems
    - ice garments
    - reflective clothing

Cold Stress

- Cold stress is a fundamentally different kind of problem than heat stress.
  - Physiological adaptations to cold stress have less dramatic effects.
- Human responses to cold stress
  - physiological
    - reduced blood circulation to skin
    - increased metabolic activity (shivering)
  - behavioral
    - increased activity
    - seeking warm locations
    - increased clothing insulation
      - insulation value
      - moisture
      - ventilation



Cold Stress Hazards

- Hazards associated with cold stress are manifested in two distinct fashions:
  - systemic (hypothermia)
    - depression of central nervous system
  - local (localized tissue damage)
    - restriction of flood flow to extremities

Note: There is also a concern for manual dexterity.

Model of Thermal Balance (simplified)

S = M + (R + C) + K – E

where: S = heat storage rate (gain [+] or loss [-])

M = metabolic rate (internal heat generation [+])

R = radiant heat exchange rate (gain [+] or loss [-])

C = convective heat exchange rate (gain [+] or loss [-])

K = conductive heat exchange (loss [-])

E = evaporative heat loss (always loss [-])

Note: The goal of systemic cold-stress control is to avoid hypothermia by limiting reduction in core temperatures to:

- 96.8°F for prolonged exposures
- 95°F for occasional, short-duration exposures

Measurement of Cold Stress

- Two climatic factors influence the rate of heat exchange between a person and the environment:
  - air temperature
  - air speed

Note: The *Wind Chill Index* was developed by the National Weather Service to reflect the effect of these variables on heat exchange.

NWS Windchill Chart

Temperature (°F)

Wind (mph) \ Calm	40	35	30	25	20	15	10	5	0	-5	-10	-15	-20	-25	-30	-35	-40	-45
5	36	31	25	19	13	7	1	-5	-11	-16	-22	-28	-34	-40	-46	-52	-57	-63
10	34	27	21	15	9	3	-4	-10	-16	-22	-28	-35	-41	-47	-53	-59	-66	-72
15	32	25	19	13	6	0	-7	-13	-19	-26	-32	-39	-45	-51	-58	-64	-71	-77
20	30	24	17	11	4	-2	-9	-15	-22	-29	-35	-42	-48	-55	-61	-68	-74	-81
25	29	23	16	9	3	-4	-11	-17	-24	-31	-37	-44	-51	-58	-64	-71	-78	-84
30	28	22	15	8	1	-5	-12	-19	-26	-33	-39	-46	-53	-60	-67	-73	-80	-87
35	28	21	14	7	0	-7	-14	-21	-27	-34	-41	-48	-55	-62	-69	-76	-82	-89
40	27	20	13	6	-1	-8	-15	-22	-29	-36	-43	-50	-57	-64	-71	-78	-84	-91
45	26	19	12	5	-2	-9	-16	-23	-30	-37	-44	-51	-58	-65	-72	-79	-86	-93
50	26	19	12	4	-3	-10	-17	-24	-31	-38	-45	-52	-60	-67	-74	-81	-88	-95
55	25	18	11	4	-3	-11	-18	-25	-32	-39	-46	-54	-61	-68	-75	-82	-89	-97
60	25	17	10	3	-4	-11	-19	-26	-33	-40	-48	-55	-62	-69	-76	-84	-91	-98

Frostbite Times: 30 minutes; 10 minutes; 5 minutes

$$\text{Wind Chill (°F)} = 35.74 + 0.6215T - 35.75(V^{0.16}) + 0.4275T(V^{0.16})$$

Where, T= Air Temperature (°F) V= Wind Speed (mph)

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Recognition of Cold Stress

- Subjective responses of workers are a good tool for recognition of cold stress in the workplace:
  - seeking warm locations
  - adding layers of clothing
  - increasing the rate of work
  - loss of manual dexterity
  - shivering
  - accidents
  - unsafe behaviors

Evaluation for Cold Stress

- Workplace monitoring
  - should be conducted when temperatures fall below 61°F

- Systemic cold stress
  - hypothermia can occur with air temperatures up to 50°F
  - ACGIH recommends employers invoke protective measures when air temperatures fall to 41°F

Control of Cold Stress

- General controls
  - training
    - additional topics for colder exposures
  - cold stress hygiene practices
    - fluid replacement
      - warm, sweet, non-caffeinated
    - diet
    - self-determination
    - wet clothing replacement
  - medical surveillance
    - medical restriction for those unable to adequately thermoregulate

Control of Cold Stress (cont.)

- Specific Controls
  - engineering controls
    - general or spot heating
    - minimizing air movement
    - reducing conductive heat transfer
    - provision of warming shelters
  - administrative controls
    - acclimation
    - work/rest cycle
    - work-sharing
    - work scheduling
    - allow for production declines
    - avoid long periods of sedentary work
    - personal monitoring
  - personal protection
    - insulated clothing
    - wind barriers
    - special attention to extremities
      - face shields
      - gloves/mittens
      - socks/footwear
    - active warming systems
    - eye protection